

NISHANTH A

+91 7397170853 · nishanthtamil72@gmail.com · [LinkedIn](#) · [github.com/Nishanthtamil](#) · [LeetCode](#) · [Portfolio](#)

Backend-focused AI engineer with hands-on experience shipping production systems — an internal AI chatbot, a ₹1,00,000 job marketplace, an AI compliance tool with hybrid vector search, and an agentic multi-language security code reviewer. Proficient in FastAPI, LangChain/LangGraph, and Milvus; experienced deploying containerised services with Docker. Seeking backend and AI engineering roles where I can contribute to scalable, real-world systems.

TECHNICAL SKILLS

Backend: FastAPI, Django, Node.js, PHP (Laravel), Next.js, REST APIs

Databases: PostgreSQL, MySQL, MongoDB, Redis, Milvus (Vector DB), Neo4j

AI / ML: LangChain, LangGraph, PyTorch, TensorFlow, vLLM, RAG pipelines, multi-agent systems

Infrastructure: Docker, AWS, Vercel, Plesk, Git, Linux, Postman, CI/CD (GitHub Actions)

Languages: Python, SQL, JavaScript, C++

Frontend: React, React Native

Concepts: Microservices, REST API design, Agile/Scrum, multi-agent systems, formal verification (Z3 SMT)

WORK EXPERIENCE

Data Science Intern — XYMA Analytics, Chennai

Jan 2025 – Mar 2025

- Built and shipped an internal team chatbot using Next.js, React, and vLLM, cutting average query resolution time by ~30% compared to manual lookups.
- Containerised the application with Docker and coordinated zero-downtime deployment on the company server alongside the networking team.
- Developed anomaly detection models for time-series sensor data achieving 96% accuracy; supported full transition from development to production.
- Designed and maintained REST API endpoints enabling structured JSON responses and session handling for 10+ concurrent internal users; deployed via CI/CD pipeline using GitHub Actions.

Freelance Software Engineer — Consultancy Project — valued at ₹1,00,000+, Remote

Oct 2024 – Dec 2024

- Architected and solo-delivered a full-scale Job Marketplace platform using PHP (Laravel) and PostgreSQL within a 3-month deadline.
- Designed a normalised database schema for listings, applications, and user roles, improving page load by ~25% over the initial design.
- Built a dynamic React UI with employer and candidate dashboards; managed production deployment and domain configuration on Plesk per client requirements.

PROJECTS

Compliance Management Chatbot — FastAPI · PostgreSQL · Milvus · Flutter — Techgium Hackathon

- Engineered a custom SOAR retrieval algorithm that reduced compliance document lookup latency by ~40% versus naive full-text search.
- Built a hybrid search backend over PostgreSQL (structured filters) and Milvus Vector DB (semantic embeddings); handled 50+ concurrent queries without degradation during hackathon judging.

Vigilant-X — Agentic Multi-Language Security Code Reviewer — Python · LangGraph · Z3 SMT · Neo4j · Docker · GitHub

- Architected an agentic PR security reviewer that builds Code Property Graphs (Joern, with clang-tidy fallback) and traces tainted data flow across file and project boundaries using Neo4j + APOC.
- Designed an LLM-to-Z3 pipeline that transpiles vulnerability patterns (use-after-free, double-free, buffer/integer overflow, command injection) into SMT constraints with temporal program-point ordering, producing formally provable findings rather than heuristic warnings.
- Built a hardened Docker sandbox that compiles proof-of-concept exploits across multiple compilers and optimisation levels and confirms crashes with LLVM sanitizers (ASan/MSan/TSan), ensuring every reported vulnerability is verified, not just suspected.
- Implemented a budget-controlled Deep Scan engine (priority-scored LLM analysis capped at 40 files/run with semaphore-based concurrency) to keep large PR reviews within LLM rate limits.
- Shipped GitHub-native inline PR review comments and SARIF 2.1.0 export for GitHub Code Scanning, supporting C/C++, Java, JavaScript/TypeScript, Python, Go, Ruby, and Rust.

EDUCATION

Rajalakshmi Institute of Technology · B.Tech in AI & Data Science — CGPA 8.2

2022 – 2026

Maharishi Vidhya Mandir MHSS, Karaikudi · Higher Secondary (Class XII) — 93.83%

2020 – 2022

Maharishi Vidhya Mandir MHSS, Karaikudi · Secondary (Class X) — 95.6%

2019 – 2020